

Λ_c^+ polarimetry with the $\Lambda_c^+ \rightarrow pK^-\pi^+$ transition

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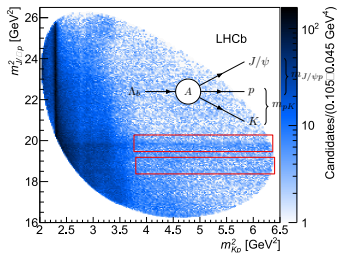
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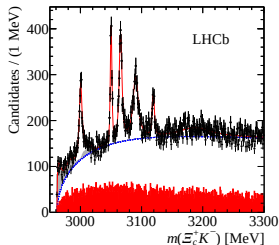
Why we want to know/measure polarization of hadrons?

- Mechanism of quark hadronization [Brambilla:2010cs, Faccioli:2010kd, Butenschoen:2012px]
- Hadron spectroscopy, extending decay chains (+ next slide)
 - ✓ $\Lambda_b^0 \rightarrow J/\psi p K$ with $J/\psi \rightarrow \mu^+ \mu^-$
 - ✓ $B \rightarrow J/\psi \bar{p} \Lambda$ with $\Lambda \rightarrow p \pi$
 - ? $B^+ \rightarrow \Lambda_c^+ \Lambda_c^- K^+$ with $\Lambda_c^+ \rightarrow p K^- \pi^+$
 - ? $\Omega_b^- \rightarrow \Xi_c^+ \pi^- K^-$ with $\Xi_c^+ \rightarrow p K^- \pi^+$
- BSM searches with $\Lambda_b^0 \rightarrow \Lambda_c^+ l^- \nu$ [Konig:1993wz, Dutta:2015ueb, Shivashankara:2015cta, Li:2016pdv, Li:2016pdv, Datta:2017aue, Ray:2018hrx, DiSalvo:2018ngq, Penalva:2019rgt, Ferrillo:2019owd]
 - E.g. sign of the longitudinal polarization of Λ_c^+ provides a test for the left-handedness of the $b \rightarrow c$ current.
- BSM searches with measurement of EDM with the charm meson

Polarization for hadron spectroscopy



- $\Lambda_b^0 \rightarrow J/\psi(\rightarrow \mu^+ \mu^-) p K^-$ [LHCb]
 - Λ_b^0 is prompt (unpolarized)
 - 2d distribution is sensitive to P_c^+ spin
 - $J/\psi \rightarrow \mu^+ \mu^-$ adds sensitivity for P_c^+ parity
- $\Xi_b^0 \rightarrow J/\psi(\rightarrow \mu^+ \mu^-) p \Lambda(\rightarrow p \pi)$ [LHCb]
 - P_{cs}^0 in $J/\psi \Lambda^0$
 - $\Lambda^0 \rightarrow p K^-$ adds sensitivity to the P_{cs} spin

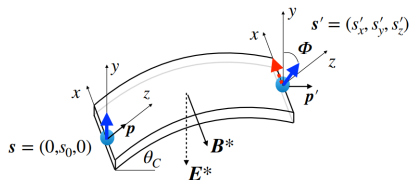


Require further input

- heavy-baryon spectroscopy: $B_b \rightarrow B_c K \pi$, with $B_c \in \{\Lambda_c^+, \Sigma_c^+, \Xi_c^+, \Omega_c^0\}$
- pentaquark searches: $\Lambda_b^0 \rightarrow \Lambda_c^+(\rightarrow p K^- \pi^+) \bar{D}^0 K^-$

Polarization for EDM studies

Search for the electric dipole moment of strange and charm baryons at LHC
(see [details here](#))



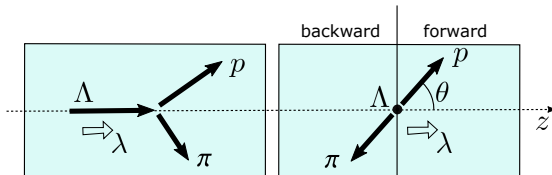
- Fixed target setup
- Production of polarized Λ_c , Ξ_c
- Propagation in the bend crystals changes orientation of the polarization

Recent publication on the progress [Aiola et al., arXiv:2010.11902]

A robust techniques to measure polarization of charm baryons is required.

Decay of fermion is special. Example $\Lambda \rightarrow p\pi^-$

Baryon in the final state averages angular distributions



$$H_\lambda = \langle p, \lambda; K^- | T_{\text{weak}} | \Lambda, \lambda \rangle : \text{hadron property}$$

$$\frac{2}{\Gamma} \frac{d\Gamma}{d\cos\theta} = 1 + P\alpha \cos\theta, \quad A_{FB} = \alpha P : \text{asymmetry}$$

- $P = |\vec{P}|$ polarization, for $J = 1/2$, there are just tree d.o.f.
- α is the asymmetry parameter (analysing power of the decay)

$$\Lambda(j^P = 1/2^+) \rightarrow p(j^P = 1/2^+) \pi^-(j^P = 0^-):$$

- S-wave – parity violating (PV); P-wave – parity conserving (PC)

$$\alpha = \frac{|H_+|^2 - |H_-|^2}{|H_+|^2 + |H_-|^2} = -\frac{2\text{Re}(H_S^* H_P)}{|H_S|^2 + |H_P|^2} \quad \text{both PV and PC are needed!}$$

Polarimetry in τ decays

[Tsai:1971vv, Kuhn:1991cc, Davier:1992nw, Kuhn:1995nn, Kuhn:1982di, Kuhn:1993ra, Hagiwara:1989fn]

$$\frac{\Phi}{\Gamma} \frac{d\Gamma}{d\Phi} \propto 1 + \vec{P} \cdot \vec{h}.$$

- $\vec{P}$ is a polarization of τ
- $\vec{h}$ is a polarimeter vector

τ polarimeter in the SM

- Unit vector, $|\vec{h}| = 1$
- direction depends on the final state
 - ▶ For $\tau^- \rightarrow \pi^- \nu_\tau$ decay, $\vec{h}$ points to the pion
 - ▶ for $\tau^- \rightarrow l^- \nu_\tau \bar{\nu}_l$ decay, $\vec{h} \uparrow \uparrow p_{\bar{\nu}}$.

Dalitz-plot decomposition

Factorization of variables describing the system

$$T_{\nu_0, \{\lambda\}}(\phi, \theta, \chi; \tau) = \sum_{\nu} D_{\nu_0, \nu}^{1/2}(\phi, \theta, \chi) A_{\nu, \{\lambda\}}(\tau)$$

Polarization d.o.f.

- Euler angles in active ZYZ convention
- Rotation of the system as rigid body
- $pp \rightarrow \Lambda_c^+ X$: (θ, ϕ) angles of Λ_c^+
- $\Lambda_c^+ \rightarrow K^* p$: (θ_K, χ) angles of K^*
- Dependence of the angles reflects polarization

Dynamic d.o.f.

- Resonances in the decay
- $m_{K\pi}$, $m_{\pi p}$, m_{pK} masses of subsystems (dependent)
- angular distribution

Polarized decay rate

$$|\mathcal{M}|^2 = \sum_{\nu_0, \nu'_0, \{\lambda\}} \Re_{\nu'_0, \nu_0} T_{\nu'_0, \{\lambda\}}^* T_{\nu_0, \{\lambda\}} ,$$

$$\Re_{\nu'_0, \nu_0} = \frac{1}{2} \left(1 + \vec{P} \cdot \vec{\sigma}^P \right)_{\nu'_0, \nu_0} ,$$

Polarized decay rate

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$$\Re_{\nu'_0, \nu_0} = \frac{1}{2} \left(1 + \vec{P} \cdot \vec{\sigma}^P \right)_{\nu'_0, \nu_0} ,$$

Note on algebra

SU(2) is double cover of SO(3):

spin 1/2, 4π rotation

The explicit homomorphism with the non-trivial center:

$$\phi : \text{SU}(2) \rightarrow \text{SO}(3), \quad \phi(d) = R ,$$

$$\phi(d) = \frac{1}{2} \text{Tr} \left[D^{1/2*}(\phi, \theta, \chi) \sigma_i^P D^{1/2}(\phi, \theta, \chi) \sigma_j^P \right] = R_{ij}(\phi, \theta, \chi),$$

Model-agnostic representation of the decay rate

Our master formula

After some gymnastic,

$$|\mathcal{M}(\phi, \theta, \chi, \tau)|^2 = I_0(\tau) \left(1 + \sum_{i,j} P_i R_{ij}(\phi, \theta, \chi) \alpha_j(\tau) \right),$$

where

- $I_0(\tau)$ is the unpolarized intensity
- $\alpha(\tau)$ is the aligned polarimeter vector,

$$\vec{\alpha}(\tau) = \sum_{\nu', \nu, \{\lambda\}} A_{\nu', \{\lambda\}}^* \vec{\sigma}_{\nu', \nu} A_{\nu, \{\lambda\}} / I_0(\tau).$$

it is specific for the decay, does not depend on the production mechanism.

- α can be computed for any model, parametrized on the grid, splines, ...

Averaging over the dynamic variables

The polarization measurement can be even simpler

Can the dynamic variables τ be integrated over, i.e. disregarded in the analysis?

$$\frac{8\pi}{\Gamma} \frac{d^3\Gamma}{d\phi d\cos\theta d\chi} = 1 + \sum_{i,j} P_i R_{ij}(\phi, \theta, \chi) \bar{\alpha}_j,$$

where

- $\vec{\alpha}$ is averaged aligned polarimeter vector
- Just three known numbers needs to be known to determine polarization
- The uncertainty on P with integrated setup is worse than with the full dimensions.

Polarization sensitivity by Davier et al.

Determination of polarization is a linear problem

$$I_0(\xi|P) = f(\xi) + Pg(\xi), \quad \text{where } \xi \text{ are kinematic variables.}$$

For the likelihood fit of the set $\{\xi_i\}_N$, the error to P can be computed analytically:

$$P = P_{\text{mod}} \pm \delta_P, \quad \delta_P = \frac{1}{S_P \sqrt{N}}, \quad \text{where } S_P^2 = \int \frac{g^2}{f + Pg} d^n \xi$$

Using the relations with our master formula

$$S_0 = 3 \int I_0 |\vec{\alpha}|^2 d^n \tau / \int I_0 d^n \tau$$

- For integrated setup, $\overline{S}_0^2 = 3(\overline{\alpha}_x^2 + \overline{\alpha}_y^2 + \overline{\alpha}_z^2)$
- The ratio $S_0/\overline{S}_0$ gives the expected increase of the statistical uncertainty

Analysis strategy

- Amplitude analysis of $\Lambda_c^+ \rightarrow pK^-\pi^+$ is finished, LHCb-PAPER-2022-002
- The model-agnostic representation will foster reusing the results in LHCb and beyond
- Implement $\Lambda_c^+ \rightarrow pK^-\pi^+$ model from LHCb-PAPER-2022-002
- Compute $\vec{\alpha}$ for every point of the $pK^-\pi^+$ Dalitz plot
- Propagate uncertainties of the angular analysis

Results:

- Four distributions I_0 , α_x , α_y , and α_z , computed on a grid of $m_{pK} \times m_{\pi K}$;
- Stat./syst. uncertainties to the maps;
- Averaged values, $\bar{\alpha}_x$, $\bar{\alpha}_y$, and $\bar{\alpha}_z$;
- Loss of precision when using $\vec{\alpha}$
- Example of application, CPV properties of $\vec{\alpha}$.

Implementation

Cross-check in two languages

ComPW_Δ

This analysis has been performed in two languages:

symbolic amplitude models



ThreeBodyDecay.jl



Both implementations have been carefully documented on an interactive webpage

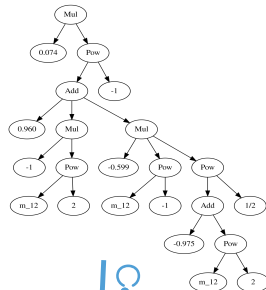
Symbolic amplitude models

COMPWA libraries

The Python implementation follows a new workflow that is facilitated by packages from the CompPWA Project (compwa-org.rtfld.io):

- 1 Formulate amplitude model **symbolically** with a Computer Algebra System
- 2 Use that symbolic expression as **template to a computational back-end**, such as a differentiable programming framework like JAX

We used JAX as the fastest backend



COMPWA



Live documentation

Self-documenting workflow

Our analysis results are automatically rendered as static webpages from Jupyter and Pluto notebooks:

[lc2pkpi-polarimetry.docs.cern.ch](https://polarimetry.docs.cern.ch)

Polarimetry $\Lambda_c \rightarrow p K \pi$

Search the docs ...

Default amplitude model

Cross-check with LHCb data

Intensity distribution

Polarimeter vector field

Uncertainties

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DPD angles

Phase space sample

Alignment consistency

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Theme by the [Executable Book Project](#)



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Spin-alignment amplitude

Sub-system amplitudes

Parameter definitions

Helicity coupling values

Non-coupling parameters

Spin-alignment amplitude

The full intensity of the amplitude model is obtained by summing the following aligned amplitude over all helicity values λ_i in the initial state 0 and final states 1, 2, 3:

```
model_choice = 0
amplitude_builder = load_model_builder(
    model_file="../../data/model-definitions.yaml",
    particle_definitions=particles,
    model_id=model_choice,
)
model = amplitude_builder.formulate()
```

> Click to show

$$\sum_{\lambda_0=-1/2}^{1/2} \sum_{\lambda_1=-1/2}^{1/2} A_{\lambda_0, \lambda_1}^1 d_{\lambda_1, \lambda_1}^{\frac{1}{2}}(\zeta_{1(1)}^1) d_{\lambda_0, \lambda_0}^{\frac{1}{2}}(\zeta_{1(1)}^0) + A_{\lambda_0, \lambda_1}^2 d_{\lambda_1, \lambda_1}^{\frac{1}{2}}(\zeta_{2(1)}^1) d_{\lambda_0, \lambda_0}^{\frac{1}{2}}(\zeta_{2(1)}^0) + A_{\lambda_1}^3$$

Note that we simplified notation here: the amplitude indices for the spinless states are not rendered and their corresponding Wigner- d alignment functions are simply 1.

The relevant $\zeta_{j(k)}^i$ angles are defined as:

> Click to show

$$\begin{aligned} \zeta_{1(1)}^0 &= 0 \\ \zeta_{1(1)}^1 &= 0 \\ \zeta_{2(1)}^0 &= -\arccos\left(\frac{-2m_0^2(-m_1^2-m_2^2+\sigma_1)+\left(m_0^2+m_1^2-\sigma_1\right)\left(m_2^2+m_3^2-\sigma_2\right)}{\sqrt{\lambda(m_0^2, m_1^2, \sigma_2)}\sqrt{\lambda(m_0^2, \sigma_1, m_1^2)}}\right) \end{aligned}$$

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> Click to show

$$\sum_{\lambda_0=-1/2}^{1/2} \sum_{\lambda_1=-1/2}^{1/2} A_{\lambda_0, \lambda_1}^1 d_{\lambda_1, \lambda_1}^{\frac{1}{2}} \left(\zeta_{1(1)}^1 \right) d_{\lambda_0, \lambda_0}^{\frac{1}{2}} \left(\zeta_{1(1)}^0 \right) + A_{\lambda_0, \lambda_1}^2 d_{\lambda_1, \lambda_1}^{\frac{1}{2}} \left(\zeta_{2(1)}^1 \right) d_{\lambda_0, \lambda_0}^{\frac{1}{2}} \left(\zeta_{2(1)}^0 \right) + A_{\lambda_1}^3$$

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The Python and Julia dependencies are pinned, so that the analysis is **fully reproducible**

Polarimetry $\Lambda_c \rightarrow p K \pi$

Search the docs ...

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Generated by the CAS

Input from LHCb analysis

Amplitude model

Implemented nominal and alternative amplitude models from LHCb's recent *Amplitude analysis of the $\Lambda_c^+ \rightarrow pK^-\pi^+$ decay* [LHCb-PAPER-2022-002]

Input from [LHCb-PAPER-2022-002]:

- the default amplitude model,
- Several alternative models with different parametrizations,
- Parameter values with error bars for each model

Present paper implements:

- Default model and alternative models, using DPD [JPAC:2019ufm] and wigner rotations
- Helicity couplings have been remapped)
- Identical dynamics lineshapes

Cross-check #1: numerical test using 2022-002 code

- Comparison for a single point in phase space
 - ▶ resonance lineshapes,
 - ▶ helicity amplitude per resonance
- agrees up to 10^{-4} .

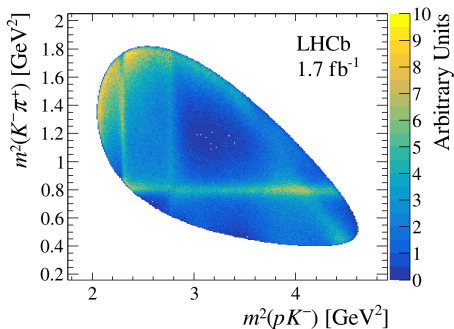
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 - amplitude

	Computed	Expected	Difference
ArD(1232)1	$\mathcal{H}_{D(1232),-\frac{1}{2},0}^{\text{production}}$		
A++	-0.488498+0.517710j	-0.488498+0.517710j	3.11e-14
A+-	0.894898-0.948412j	0.894898-0.948412j	7.61e-15
A-+	0.121490-0.128755j	0.121490-0.128755j	1.80e-14
A--	-0.222563+0.235872j	-0.222563+0.235872j	6.14e-15
ArD(1232)2	$\mathcal{H}_{D(1232),\frac{1}{2},0}^{\text{production}}$		
A++	-0.222563+0.235872j	-0.222563+0.235872j	6.14e-15
A+-	-0.121490+0.128755j	-0.121490+0.128755j	1.80e-14
A-+	-0.894898+0.948412j	-0.894898+0.948412j	7.61e-15
A--	-0.488498+0.517710j	-0.488498+0.517710j	3.11e-14
ArD(1600)1	$\mathcal{H}_{D(1600),-\frac{1}{2},0}^{\text{production}}$		

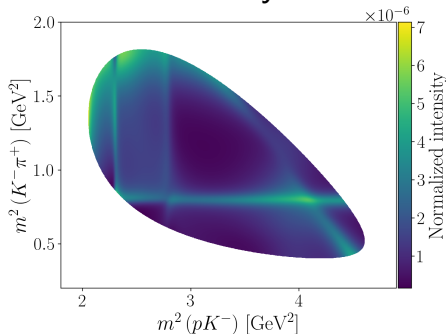
- Contents
- Lineshape comparison
- Amplitude comparison**
- SympPy expressions
- Numerical functions
- Input data
- Comparison table

Cross-check #2: visual similarity

LHCb-PAPER-2022-002

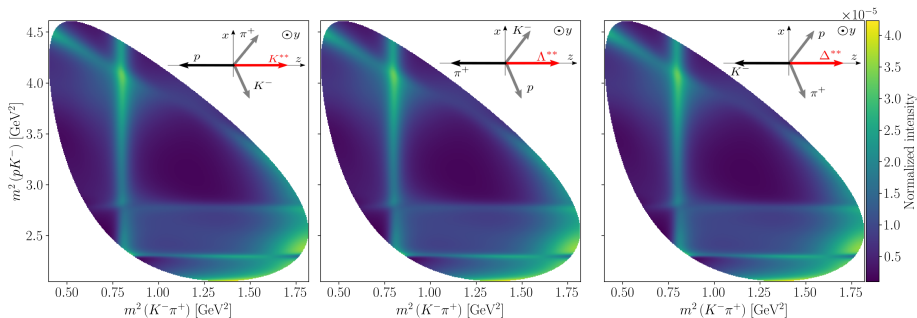


This analysis



Cross-check #3: choice of alignment

Intensity distributions are exactly the same for each DPD alignment



Polarimeter field in Dalitz coordinates

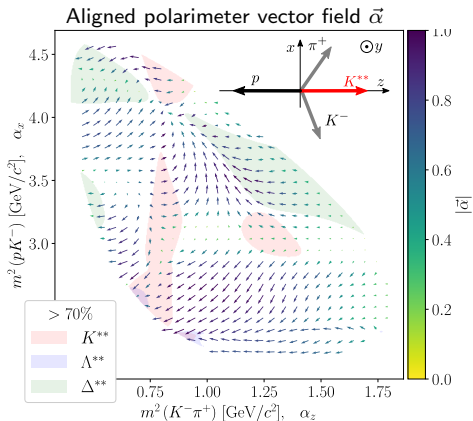
Amplitude to polarimeter:

$$\vec{\alpha}(\tau) = \sum_{\nu', \nu, \{\lambda\}} A_{\nu', \{\lambda\}}^* \vec{\sigma}_{\nu', \nu} A_{\nu, \{\lambda\}} / I_0$$

Polarimeter to intensity:

$$|\mathcal{M}(\phi, \theta, \chi, \tau)|^2 = I_0(\tau) \times \left(1 + \sum_{ij} P_i R_{ij}(\phi, \theta, \chi) \alpha_j(\tau) \right)$$

$$R(\phi, \theta, \chi) = R_Z(\phi) R_Y(\theta) R_Z(\chi)$$

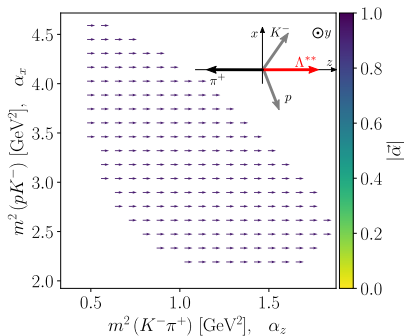


Understanding the polarimetry map

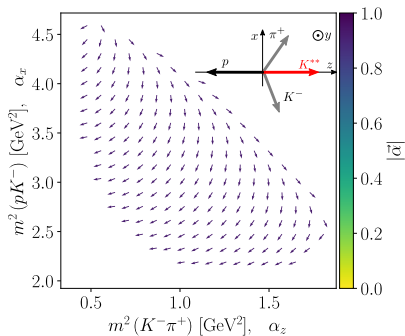
Example: $\Lambda_c^+ \rightarrow \Lambda(1520)(\rightarrow pK^-)\pi^+$

$\vec{\alpha}$ of individual contributions points in z-direction
when the resonance is aligned with z.

Aligned



Misaligned



Propagation of the uncertainties

Uncertainties over the polarimeter field have two components:

- **statistical:**

- ▶ parameter **resampling** of the nominal model using parameter error bars
- ▶ RMS over the resulting parameter-resampled distributions

- **model:**

- ▶ implement alternative models
- ▶ only central values of alternative models are considered
- ▶ take min-max extrema

- **systematic:**

- ▶ Small compare to stat.&model
- ▶ included in the **resampling**.

Table 8: Default amplitude model measured fit parameters describing the A contributions.

Parameter	Central Value	Stat. Unc.	Model Unc.	Syst. Unc.
$\text{Re}H_{1/2,0}^{A(1405)}$	-4.6	0.5	3.3	0.1
$\text{Im}H_{1/2,0}^{A(1405)}$	3.2	0.5	3.2	0.1
$\text{Re}H_{-1/2,0}^{A(1405)}$	10	1	12	0
$\text{Im}H_{-1/2,0}^{A(1405)}$	2.8	1.1	3.7	0.3
$\text{Re}H_{1/2,0}^{A(1520)}$	0.29	0.05	0.12	0.01
$\text{Im}H_{1/2,0}^{A(1520)}$	0.04	0.05	0.12	0.02
$\text{Re}H_{-1/2,0}^{A(1520)}$	-0.16	0.14	0.69	0.03
$\text{Im}H_{-1/2,0}^{A(1520)}$	1.5	0.1	1.3	0.0
$m^{A(1520)} [\text{MeV}]$	1518.47	0.36	0.65	0.03
$\Gamma^{A(1520)} [\text{MeV}]$	15.2	0.8	1.3	0.1
$\text{Re}H_{1/2,0}^{A(1600)}$	4.8	0.5	5.0	0.1
$\text{Im}H_{1/2,0}^{A(1600)}$	3.1	0.5	3.7	0.1
$\text{Re}H_{-1/2,0}^{A(1600)}$	-7.0	0.5	8.7	0.1
$\text{Im}H_{-1/2,0}^{A(1600)}$	0.8	0.6	2.0	0.2
$\text{Re}H_{1/2,0}^{A(1670)}$	-0.34	0.05	0.35	0.01
$\text{Im}H_{1/2,0}^{A(1670)}$	-0.14	0.05	0.22	0.02
$\text{Re}H_{-1/2,0}^{A(1670)}$	-0.57	0.10	0.46	0.02
$\text{Im}H_{-1/2,0}^{A(1670)}$	1.0	0.1	1.2	0.0
$\text{Re}H_{1/2,0}^{A(1690)}$	-0.39	0.10	0.23	0.02
$\text{Im}H_{1/2,0}^{A(1690)}$	-0.11	0.09	0.44	0.02
$\text{Re}H_{-1/2,0}^{A(1690)}$	-2.7	0.2	2.4	0.0
$\text{Im}H_{-1/2,0}^{A(1690)}$	-0.35	0.23	0.60	0.06
$\text{Re}H_{1/2,0}^{A(2000)}$	-8	1	11	0
$\text{Im}H_{1/2,0}^{A(2000)}$	-7.6	0.8	7.7	0.2
$\text{Re}H_{-1/2,0}^{A(2000)}$	-4.3	0.5	3.4	0.2
$\text{Im}H_{-1/2,0}^{A(2000)}$	-3.8	0.4	3.7	0.1
$m^{A(2000)} [\text{MeV}]$	1988	2	21	1
$\Gamma^{A(2000)} [\text{MeV}]$	179	4	16	3

Uncertainties on the polarimetry field

We compute $\vec{\alpha}_i$ over a phase space sample, with i one of the parameter resamplings of the 17 models or .

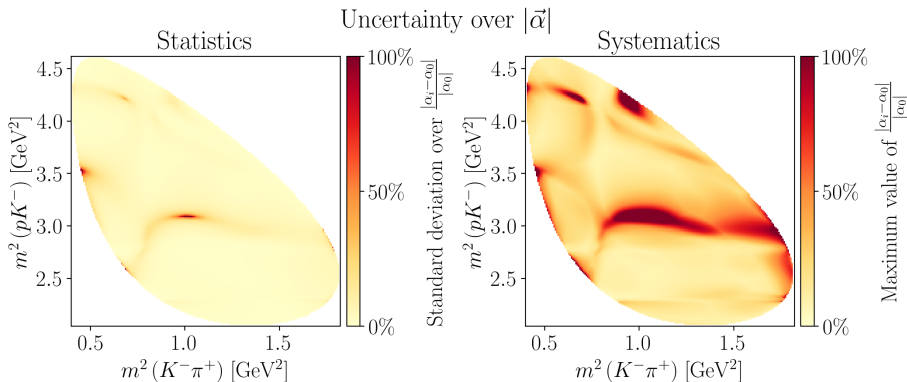
Uncertainties are then visualized separately:

- **vector norm:** $\frac{\alpha_i - \alpha_0}{\alpha_0}$
- **solid angle:**

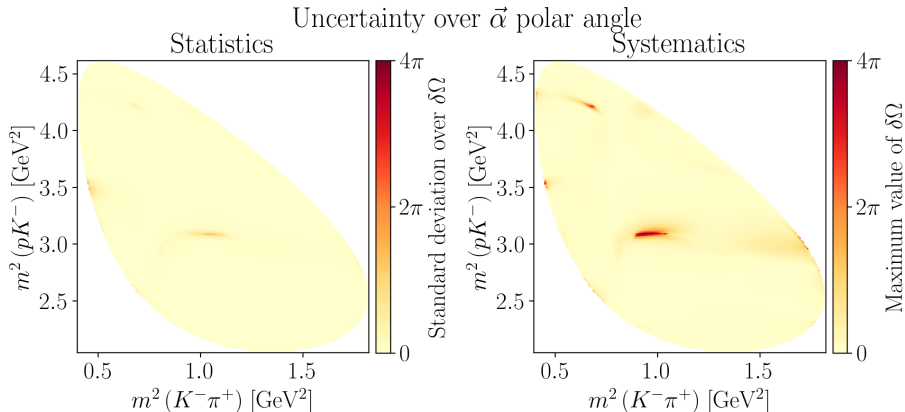
$$\delta\Omega = \int_0^{2\pi} \int_0^\theta d\phi d\cos\theta = 2\pi(1 - \cos\theta).$$

with $\cos\theta_i = \frac{\vec{\alpha}_i \cdot \vec{\alpha}_0}{|\alpha_i||\alpha_0|}$.

Uncertainties on the polarimetry field



Uncertainties on the polarimetry field



Averaged polarimeter field

$$\bar{\alpha}_j = \int l_0 \alpha_j d^n \tau / \int l_0 d^n \tau .$$

This results in:

$$\begin{aligned} \overline{\alpha_x} &= (-62.6 \pm 4.5^{+8.4}_{-14.8}) \times 10^{-3}, \\ \overline{\alpha_y} &= (+8.9 \pm 8.9^{+9.1}_{-12.7}) \times 10^{-3}, \quad (\text{due to interf.}) \\ \overline{\alpha_z} &= (-278.0 \pm 23.7^{+12.6}_{-40.4}) \times 10^{-3}, \\ |\overline{\alpha}| &= (669.4 \pm 9.3^{+15.3}_{-10.4}) \times 10^{-3}, \quad (\approx |\overline{\alpha}| \times 3) \end{aligned}$$

First uncertainty is stat.&syst (standard deviation),
second is model (extrema of the alternative models)

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First uncertainty is stat.&syst (standard deviation),
second is model (extrema of the alternative models)

Spherical coordinates (less correlated resampling uncertainty):

$$\begin{aligned}|\overline{\alpha}| &= (+285.1 \pm 24.0^{+37.9}_{-13.8}) \times 10^{-3}, \\ \theta(\overline{\alpha}) &= (+0.929 \pm 0.002^{+0.017}_{-0.011}) \times \pi, \quad (\text{small error!}) \\ \phi(\overline{\alpha}) &= (+0.955 \pm 0.045^{+0.067}_{-0.028}) \times \pi,\end{aligned}$$

Data availability

Computed $\vec{\alpha}$ fields are available on [our website](#) in JSON format over a 100x100 Dalitz plot grid. What we offer:

- the nominal model and all alternative models
- 100 parameter resamplings of the nominal model

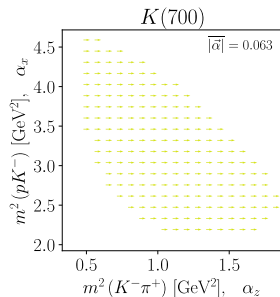
This makes it possible to **propagate statistical and systematic uncertainties** in further analyses that use the $\Lambda_c^+ \rightarrow pK^-\pi^+$ polarimetry field

Summary

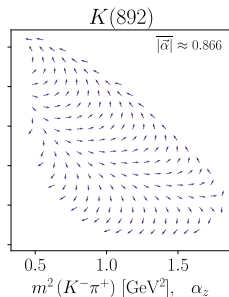
- New result: computed **baryon polarimetry field for a three-body decay** from recent $\Lambda_c^+ \rightarrow pK^-\pi^+$ amplitude analysis by LHCb
- New approach: **symbolic amplitude models** with JAX as computational back-end
- Computed polarimeter field is offered for re-use in other analyses involving Λ_c^+

Appendix: $\vec{\alpha}$ for baryon + vector

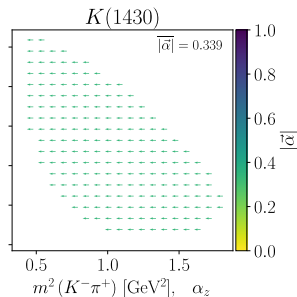
Example: $\Lambda_c^+ \rightarrow K^{**}(\rightarrow \pi^+ K^-) p$ aligned to p



$$1/2^\pm \rightarrow 1/2^+ \otimes 0^-$$



$$1/2^\pm \rightarrow 1/2^+ \otimes 1^-$$



$$1/2^\pm \rightarrow 1/2^- \otimes 0^-$$

More appendices

- Use of the polarimetry in $B^+ \rightarrow \Lambda_c^+ \bar{\Lambda}_c^- K^+$
- CP conjugation of the polarimetry field

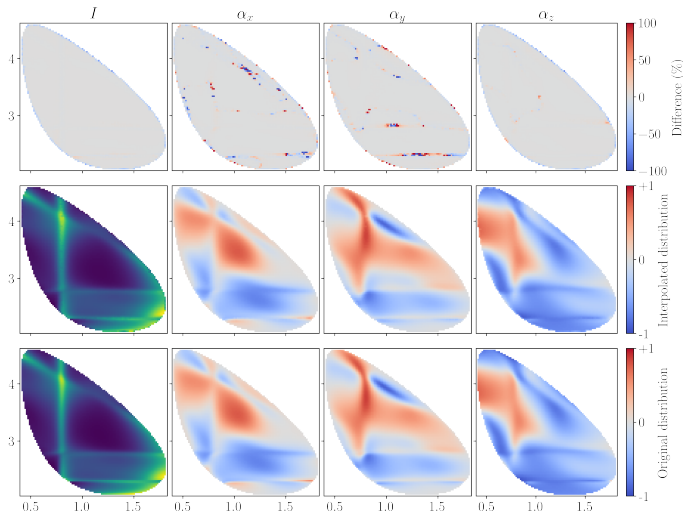
Discretization error

Justification of the 100^2 grid

The $\alpha_i(\tau)$ are exported on grid:

- The grid size in $\delta m_{pK} \sim \Gamma_{\Lambda(1520)}$
- Toys of $\vec{P}$ fits with 100^2 , 200^2 , 500^2 , exact $\Rightarrow$ negligible extra uncertainty
- ~ 70 Mb output json

Comparison interpolated and actual values



Thanks

Many thanks to

- RC: Liming Zhang, Yiming Li
- EB reader: Diego Milanes
- Management: Yasmine, Carla
- Proponents of LHCb-PAPER-2022-002: Daniele, Nicola, Fernando
- Charm WG conveners: Marianna, Dominik

Analysis information

Twiki page

<https://twiki.cern.ch/twiki/bin/view/LHCbPhysics/PolarimetryLc2pKpi>

Analysis documentation

<https://lc2pkpi-polarimetry.docs.cern.ch/index.html>

Paper draft

[\[twiki link\]](#)

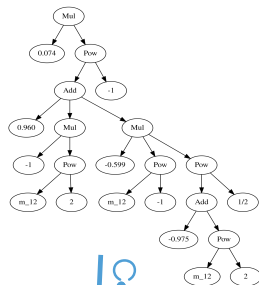
Backup slides

Computation backend

From symbolic tries to fast compiled code

Advantages of this workflow:

- Computational implementation is **outsourced to fast, optimized back-ends** from the Machine Learning and data science community
- Out-of-the-box GPU and multithreading support
- Very easy to implement other parametrizations without having to worry about performance
- Symbolic amplitude models result in a **self-documenting workflow**
- CAS simplifications result in performance boosts



COMPWA

